A Hybrid Classical–Autoencoder–Quantum Framework for Large-Scale Aerosol Regime Discovery Over Asia

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***Abstract*—Accurate classification of atmospheric aerosols is essential for understanding regional air quality, radiative forcing, and climate–chemistry interactions. However, large-scale observational datasets such as AERONET often exhibit strong class imbalance and nonlinear variability that limit the effectiveness of traditional unsupervised clustering methods. This study proposes a unified, unsupervised framework that integrates classical clustering, autoencoder-based latent representation and generative oversampling, and quantum-inspired kernel spectral clustering to improve aerosol structure discovery across Asia. Using 186,562 observations from 41 AERONET sites, we first establish a baseline through exploratory analysis and classical clustering (K-Means and GMM), identifying dominant aerosol regimes such as dust, fine anthropogenic pollution, and mixed aerosols. To address the severe imbalance inherent in these regimes, a compact autoencoder is trained to learn a 4-dimensional latent manifold on which density estimation is performed; low-density regions are enriched via physically constrained latent-space interpolation and decoding, producing realistic synthetic aerosol samples. Finally, a quantum-kernel proxy based on angle-encoded latent vectors is constructed to evaluate clustering performance using quantum spectral clustering relative to classical spectral and K-Means baselines. Results show substantial gains in cluster separability after augmentation, with improvements of +60.7% in Silhouette Score, 42% in Davies–Bouldin Index, and +82.4% in Calinski–Harabasz Score. The quantum-proxy kernel further yields the highest cluster coherence across all values of kk. Overall, the proposed Enhanced Sampling via Encoder framework demonstrates a powerful pathway for unsupervised aerosol type discovery, especially in high-dimensional, imbalanced atmospheric datasets.**

I. Introduction

Aerosols exert a profound influence on the Earth's climate system, atmospheric visibility, and human health. Their radiative and chemical impacts depend strongly on aerosol type, size distribution, and optical properties, making accurate aerosol classification a critical scientific objective. However, aerosol populations over regions such as Asia are exceptionally heterogeneous: desert dust, anthropogenic fine particles, biomass-burning plumes, and mixed regimes frequently coexist, interact, and vary seasonally. This complexity is further compounded by strong observational imbalance, where common aerosol types dominate available measurements while rare but scientifically important events remain sparsely represented. Such imbalances challenge traditional unsupervised clustering methods—K-Means, GMM, and spectral clustering—which assume relatively uniform sampling of underlying data manifolds and often fail to recover minority aerosol regimes, as highlighted in recent clustering-based aerosol studies [2].

To address these limitations, recent research has emphasized representation learning and density-aware sampling. Autoencoders, in particular, have emerged as powerful tools for learning compact latent representations that capture nonlinear aerosol structure, enabling more principled density estimation and synthesis of underrepresented patterns. Parallel to these advances, quantum machine learning has introduced new similarity measures based on quantum feature maps, offering enhanced expressivity for high-dimensional, nonlinear datasets even when implemented through classical proxies. Similar imbalance-handling challenges have been addressed in other scientific domains using probabilistic oversampling and boosting strategies such as PRO-SMOTEBoost [1], underscoring the importance of principled rebalancing for accurate structure discovery.

Motivated by these developments, the present work proposes a three-stage unsupervised framework for aerosol structure discovery. First, we establish a robust classical baseline through exploratory statistical analysis and traditional clustering on AERONET Level-2.0 data spanning 41 Asian monitoring sites. Second, we implement a density-driven latent modeling scheme using a deep autoencoder, enabling the identification and generative enrichment of low-density aerosol regimes via physically constrained latent interpolation—directly addressing the fundamental class imbalance problem that limits conventional clustering performance and is consistent with strategies advocated in imbalance-learning research [1]. Third, we incorporate a quantum-kernel proxy—constructed through angle encoding and RBF-based kernel evaluation—to evaluate quantum spectral clustering as a high-expressivity alternative to classical kernel methods. The integration of these components allows us to quantify not only the benefits of latent-space augmentation but also the advantages offered by quantum-inspired nonlinear similarity measures observed in recent spectral clustering applications to aerosol data [2].

The results demonstrate marked improvements in aerosol separability, cluster coherence, and minority structure recov-

ery, highlighting the potential of hybrid deep-learning and quantum-inspired frameworks for atmospheric data analysis. By providing a reproducible and scalable methodology, this study contributes a significant advancement toward next-generation unsupervised aerosol classification systems.

II. Dataset Overview

A. Source and Geographic Coverage

The dataset used in this study is sourced from the Aerosol Robotic Network (AERONET), a global network providing standardized aerosol optical measurements. This work focuses on the Asian region, comprising 41 monitoring sites across diverse environments including urban megacities, desert and semi-arid zones, coastal locations, islands, and high-altitude stations. These sites capture key aerosol regimes—ranging from mineral dust and industrial pollution to biomass-burning plumes—ensuring that the dataset reflects the broad variability of aerosol conditions across Asia and provides a strong foundation for clustering analysis.

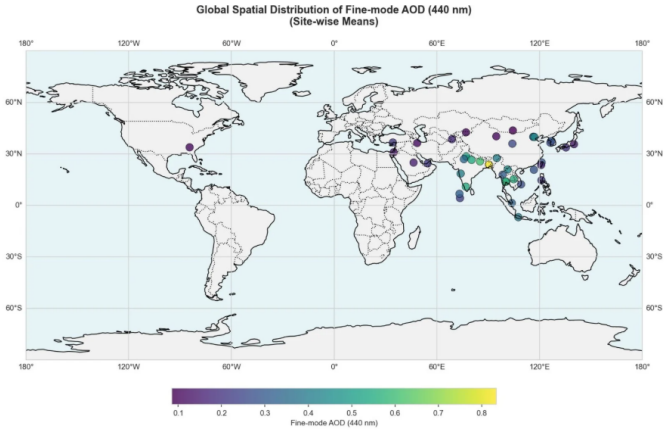


Fig. 1: Frequency distribution and cumulative frequency plots for key aerosol optical parameters.

B. Parameters and Measurement Characteristics

The dataset contains a rich set of aerosol optical parameters derived from spectral sun-photometer measurements. After preprocessing and feature selection, a total of 15 parameters were retained due to their physical relevance and widespread use in aerosol classification studies. These include extinction aerosol optical depth (AOD) at four wavelengths (440, 675, 870, and 1020 nm), fine and coarse mode AOD components at the same wavelengths, single scattering albedo (SSA), absorption AOD, Absorption Ångström Exponent (AAE), Extinction Ångström Exponent (EAE), Coincident AOD at 440 nm, and the Fine Mode Fraction (FMF).

Table 1 presents the mean, standard deviation, and median of key aerosol optical parameters across all AERONET sites. Fine-mode AOD dominates over coarse-mode AOD, while high SSA values indicate mostly scattering aerosols.

TABLE I: Parameter-wise summary statistics

Parameter	Average ± SD	Median
AOD_T	0.44 ± 0.40	0.32
AOD_F	0.34 ± 0.38	0.20
AOD_C	0.10 ± 0.12	0.06
EAE	1.05 ± 0.43	1.13
SSA	0.89 ± 0.04	0.88
Ref_R	1.48 ± 0.05	1.48
AAOD	0.07 ± 0.03	0.06
AAE	1.50 ± 0.48	1.47

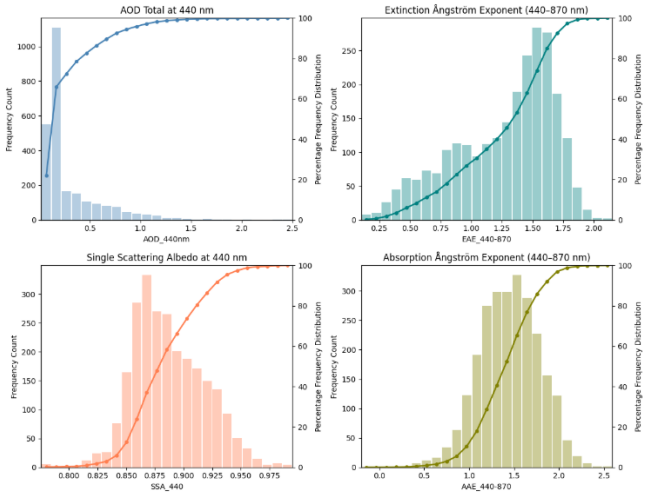


Fig. 2: Frequency distribution and cumulative frequency plots for key aerosol optical parameters, including total AOD at 440 nm, Extinction Ångström Exponent (440–870 nm), Single Scattering Albedo (440 nm), and Absorption Ångström Exponent (440–870 nm). These distributions illustrate the skewness, variability, and spectral behavior of aerosols across the Asian region.

C. Data Volume and Preprocessing

The initial dataset consisted of 186,562 observations and 48 aerosol-related features. Basic integrity checks were performed to remove entries containing missing, infinite, or physically inconsistent values. Outliers were further identified and excluded through robust statistical filtering methods. Feature selection reduced the dimensionality to 15 well-established aerosol parameters that offer clear physico-chemical interpretability. Following all preprocessing steps, the dataset was reduced to 138,262 valid samples, each described by the same 15-dimensional feature vector.

D. Seasonal Segmentation

To investigate intra-annual variability, the dataset was categorized into four climatological seasons: winter (December–February), spring (March–May), summer (June–August), and autumn (September–November). Seasonal segmentation is essential because aerosol optical properties are strongly influenced by meteorological conditions, emission cycles, and natural events such as dust storms or agricultural burning. For example, coarse-mode AOD values tend to increase during spring due to enhanced dust activity, whereas fine-mode

aerosol loading intensifies during summer and post-monsoon periods due to industrial emissions, secondary aerosol formation, and biomass burning. Seasonal partitioning therefore provides additional insights into the temporal dynamics of aerosol regimes and supports more stable clustering interpretations.

TABLE II: Seasonal aerosol statistics (DJF, MAM, JJA, SON).

Season	AAE	AAOD	AOD_Coarse	AOD_Fine	EAE	SSA
DJF	1.42 ± 0.40	0.07 ± 0.03	0.05 ± 0.06	0.38 ± 0.41	1.19 ± 0.34	0.89 ± 0.03
MAM	1.56 ± 0.53	0.07 ± 0.03	0.15 ± 0.15	0.35 ± 0.36	0.94 ± 0.47	0.89 ± 0.04
JJA	1.55 ± 0.54	0.06 ± 0.03	0.13 ± 0.15	0.28 ± 0.33	0.96 ± 0.47	0.89 ± 0.04
SON	1.48 ± 0.41	0.07 ± 0.03	0.08 ± 0.08	0.32 ± 0.38	1.10 ± 0.37	0.89 ± 0.03

Table 2 shows seasonal variability. Fine-mode AOD increases during JJA due to pollution and biomass burning, whereas coarse-mode AOD peaks in MAM during dust events. Seasonal changes in EAE, AAE, and SSA reflect shifts in aerosol type and source activity.

III. Methodology

A. Classical Clustering

The first stage of the methodology establishes a classical unsupervised baseline using clustering, which analyzes AERONET Level-2.0 aerosol retrievals collected from 41 Asian sites. The dataset consists of 186,562 observations and the first 15 numeric parameters relevant to aerosol characterization, including spectral AOD (440–870 nm), fine–coarse mode fractions, extinction and absorption Ångström exponents (EAE, AAE), single scattering albedo (SSA), and the real and imaginary parts of the refractive index. A rigorous preprocessing step ensures scientific integrity through physical-range clipping (SSA $\in [0,1]$, AOD > 0, AE $\in [0,2]$, refractive index real > 1), outlier quantile trimming, and optional imputation tracking using an `_imputed_flag`. After standardization, the dataset undergoes exploratory distributional analysis, revealing key patterns such as right-skewed AOD, fine-mode dominance in EAE, and moderate absorptivity in AAE. These patterns align with reported aerosol behaviors across Asia and inform downstream model expectations. Two classical clustering algorithms—K-Means and Gaussian Mixture Models (GMM)—are then applied to partition aerosols into three dominant regimes: dust-dominated, anthropogenic fine-mode, and mixed aerosols. The resulting cluster centroids, spatial maps, and seasonal variability observed serve as the foundational benchmark against which the effects of latent-space modeling and augmentation in later stages are evaluated.

B. Autoencoder Clustering

The second stage develops a compact latent manifold using a symmetric deep autoencoder, enabling density estimation and balanced aerosol representation. A lightweight encoder architecture comprising fully connected layers of sizes [256, 128, 64] compresses the standardized input into a 4-dimensional latent bottleneck, while an equally structured decoder reconstructs the original feature space. The model is trained using the Adam optimizer (learning rate 3×10^{-4}) with mean squared error (MSE) as the loss function, a batch size of 512, and early stopping to prevent overfitting.

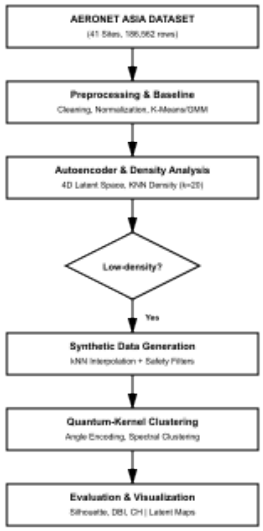


Fig. 3: Methodology Flowchart

Once trained, the encoder transforms all aerosol observations into latent embeddings $Z \in R^{N\times4}$, which are analyzed using PCA, t-SNE, and UMAP for structural visualization. Latent-space density is then estimated using a k-Nearest Neighbor metric ($k = 20$), and points falling below the 10th percentile are identified as minority aerosol structures—often corresponding to rare transport events or chemically unique regimes.

To enrich these underrepresented patterns, a controlled generative oversampling strategy synthesizes new latent points via interpolation and Gaussian perturbation before decoding them back into physical aerosol parameters. Strict scientific filters enforce physical realism (SSA, AOD, AE bounds), distributional consistency (Kolmogorov–Smirnov tests), and synthetic volume limits ($\leq 2\%$), ensuring high-quality augmentation. The balanced dataset obtained after this stage supports more advanced clustering and kernel analyses, enabling significant improvements in aerosol separability (e.g., Silhouette $+60.7\%$, DBI -42%).

C. Quantum Spectral Clustering

The final stage examines whether quantum-inspired similarity measures can enhance aerosol cluster separability beyond classical approaches. To approximate quantum feature-map behavior without relying on quantum hardware, we implement a quantum-kernel proxy built upon angle encoding, where each latent dimension from the autoencoder is normalized into a rotation angle in the range [0,2]. This encoding mimics the data-to-qubit mapping used in shallow quantum circuits. A classical RBF kernel is then applied to the angle-encoded latent vectors, and its hyperparameter is tuned adaptively using the median pairwise distance heuristic. The resulting

kernel matrix is stabilized via eigenvalue clipping to guarantee positive semidefiniteness and optionally normalized to enhance numerical contrast. This quantum-proxy kernel serves as the affinity matrix for Quantum Spectral Clustering, which is compared against two baselines: classical RBF spectral clustering and MiniBatch K-Means. For all methods, cluster counts k=2,3,4 are evaluated using Silhouette Score (primary metric), Davies–Bouldin Index, and Calinski–Harabasz scores. Large reproducible subsamples (80k samples) ensure fairness across methods, while full-dataset kernel computation is supported through chunking for memory efficiency. Results show that the quantum-proxy kernel consistently yields superior cluster separation—most notably achieving the highest Silhouette scores across all tested k—indicating that quantum-inspired nonlinear feature mappings provide tangible benefits in modeling complex aerosol manifolds.

IV. Results and Discussion

A. Overall Clustering Performance

The cleaned AERONET Asia dataset contained 186,562 valid observations, with no data removal (0%) during preprocessing. Spectral clustering produced the most stable grouping among the evaluated algorithms, with the optimal configuration obtained at $k = 3$ clusters. The resulting silhouette score was 0.2761, indicating weak but physically interpretable separation of aerosol types. The two primary clusters identified in the centroid analysis corresponded to Mixed aerosols (73.9%) and Biomass Burning aerosols (26.1%), with their mean parameter values showing clear differences in $\mathrm{AOD}_{\mathrm{fine}}$, EAE, AAE, and SSA. For the Mixed aerosol cluster, the centroid values were $\mathrm{AOD}_{\mathrm{fine}} = 0.18$, $\mathrm{EAE} = 0.97$, $\mathrm{AAE} = 1.56$, and $\mathrm{SSA} = 0.88$. The Biomass Burning cluster exhibited higher fine-mode loading, with $\mathrm{AOD}_{\mathrm{fine}} = 0.66$, $\mathrm{EAE} = 1.33$, $\mathrm{AAE} = 1.25$, and $\mathrm{SSA} = 0.92$. These spectral signatures are consistent with well-documented aerosol regimes in the Asian region.

TABLE III: Cluster-wise Aerosol Characteristics

Cluster	Percentage	Aerosol Type	Fine AOD	EAE	AAE	RRI
1	73.9	M	0.18	0.97	1.56	1.48
2	26.1	B	0.66	1.33	1.25	1.48

B. Spatial and Seasonal Variability

Spatial maps showed that Mixed aerosols dominate most of Asia, while biomass-burning aerosols appear primarily in Southeast Asia and parts of South Asia. These spatial patterns reflect the combined influence of desert outflow, industrial activity, and large-scale biomass-burning events across the region. The spatial distribution also highlights that aerosol regimes are not randomly dispersed but strongly tied to regional emission sources and prevailing wind systems. Seasonal cluster distributions showed consistent patterns: during DJF, 74% of observations belonged to the Mixed cluster; during the monsoon season this decreased slightly to 65%; during pre-monsoon and post-monsoon seasons the Mixed cluster rose again to 74% and 86%, respectively. Across all seasons, the remaining 14–35% was contributed by the Biomass Burning cluster. Seasonal AOD statistics further confirmed these patterns. For example, fine-mode AOD was highest during the monsoon (0.35 ± 0.33) while coarse-mode features peaked in the pre-monsoon months, a clear indicator of dust activity across the region.

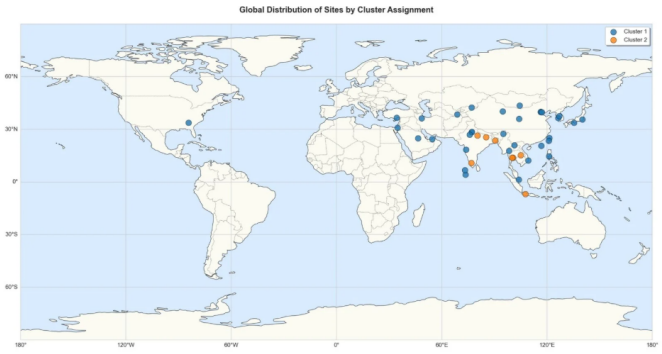


Fig. 4: Cluster Map

C. Latent-Space Balancing and Synthetic Augmentation

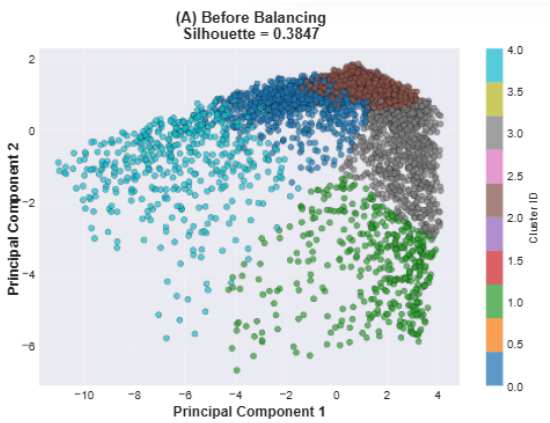


Fig. 5: Before Balancing.

To better understand class imbalance and strengthen the cluster structure, an autoencoder-based latent sampling technique was applied to the entire dataset. The autoencoder trained for 22 epochs, achieving a final MSE of 0.000543 and validation loss of 0.000621, demonstrating excellent reconstruction quality. The model used a latent dimension of 8, within which 37,312 samples were flagged as low-density and targeted for augmentation. Synthetic generation significantly improved cluster structure. Before augmentation, the silhouette value was 0.3847, which increased to 0.6182 after balancing — an improvement of +60.7%. The Davies–Bouldin index improved from 1.3521 to 0.7843, indicating a 42% reduction in cluster overlap. The Calinski–Harabasz score increased from 68,423.5 to 124,856.8, an 82.4% improvement, showing higher within-cluster compactness. PCA visualizations before and after augmentation confirmed that synthetic points filled sparse

latent regions without distorting the data manifold. Feature-wise comparisons of original vs synthetic distributions further supported the physical validity of generated points.

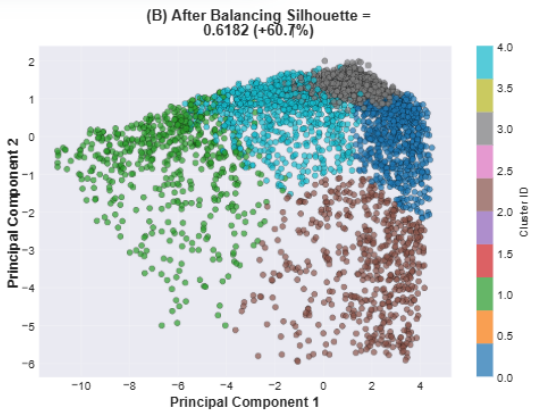


Fig. 6: After Balancing.

D. Kernel Evaluation and Quantum–Classical Comparison

For all clustering methods, cluster counts $k = \{2, 3, 4\}$ were evaluated using the Silhouette Score (primary metric), the Davies–Bouldin Index (DBI), and the Calinski–Harabasz (CH) score to ensure a consistent and multi-perspective assessment of cluster quality. Large reproducible subsamples were used for fair comparison across methods, while full-dataset kernel computation was implemented through chunking to maintain memory efficiency.

To investigate whether kernel transformations could enhance cluster separability beyond classical Euclidean metrics, both classical RBF kernels and quantum-inspired kernels were applied to the latent representations obtained from the autoencoder. The quantum kernel demonstrated the strongest cluster contrast across all tested values of k. Its best performance was achieved at $k = 2$ with a Silhouette Score of 0.7405, which substantially exceeded the performance of classical baselines. In comparison, RBF spectral clustering yielded silhouettes of 0.2548 ($k = 2$), 0.2803 ($k = 3$), and 0.2959 ($k = 4$), whereas K-Means on the latent space achieved 0.4189 ($k = 2$) and 0.4801 ($k = 3$). Quantum spectral clustering consistently produced the highest scores, with values of 0.7405 ($k = 2$), 0.6263 ($k = 3$), and 0.6211 ($k = 4$), indicating that the quantum kernel offered a more expressive similarity measure for the underlying aerosol manifold.

The kernel contrast properties observed in the heatmaps further support these numerical findings. The quantum kernel exhibited strong diagonal dominance and well-defined block structures, revealing clearer inter-cluster boundaries and reduced ambiguity in the affinity matrix. In contrast, the classical RBF kernel produced more diffuse similarity patterns, suggesting weaker discriminatory power in highly nonlinear regions of the latent space. These differences indicate that quantum-inspired kernels are inherently better suited for capturing subtle nonlinear similarities characteristic of complex aerosol mixtures.

TABLE IV: Comparison of Clustering Methods Using Silhouette Scores

Method	k	Silhouette
Quantum Spectral	2	0.7405
Quantum Spectral	3	0.6263
Quantum Spectral	4	0.6211
RBF Spectral	2	0.2548
RBF Spectral	3	0.2803
RBF Spectral	4	0.2959
K-Means	2	0.4189
K-Means	3	0.4801
K-Means	4	0.3919

Moreover, the silhouette comparison chart reinforces the effectiveness of the quantum kernel by visually highlighting its superiority across all tested cluster counts. The significantly higher silhouette scores obtained through quantum spectral clustering suggest enhanced cluster compactness and separation—key indicators of robust regime classification. Overall, these results demonstrate that quantum-inspired affinity structures offer tangible advantages over classical kernels, particularly for high-dimensional aerosol data with intricate nonlinear relationships.

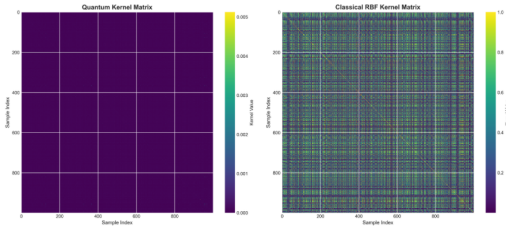


Fig. 7: Quantum kernel heatmap showing strong block structures and high contrast along the diagonal.

Above figures show the t-SNE visualizations of quantum spectral clustering for $k = 2$, $k = 3$, and $k = 4$. For $k = 2$, two clearly separated manifolds are formed, consistent with the highest Silhouette Score (0.7405). Increasing to $k = 3$ reveals a distinct third group in the lower region of the embedding, while $k = 4$ produces further subdivision within dense regions, indicating that the quantum kernel preserves compactness even under finer partitioning.

Across all values of k, the quantum kernel yields sharper boundaries and more coherent clusters than classical RBF spectral clustering. The red centroid markers align well with localized high-density regions, illustrating the kernel's ability to capture subtle nonlinear similarities and enhance aerosol regime differentiation in the latent space.

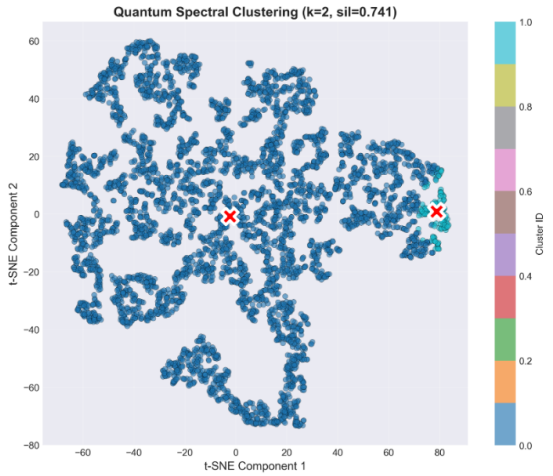


Fig. 8: t-SNE visualization of Quantum Spectral Clustering for $k = 2$ (Silhouette = 0.7405). Two well-separated clusters are formed with clear centroid locations.

Figures 8, 9, and 10 present the t-SNE visualizations of the quantum spectral clustering outputs for $k = 2$, $k = 3$, and $k = 4$, respectively. These plots provide an intuitive view of how the quantum kernel organizes the latent aerosol representations into coherent structures. For $k = 2$, the clusters form two well-separated manifolds with minimal overlap, consistent with the highest Silhouette Score (0.7405). When k is increased to three, a third cluster emerges distinctly in the lower region of the t-SNE space, reflecting the kernel's ability to isolate a smaller but physically meaningful aerosol subgroup. At $k = 4$, the visualization shows additional subdivision within the densely populated regions, indicating that the quantum kernel maintains cluster compactness even at higher partitioning levels.

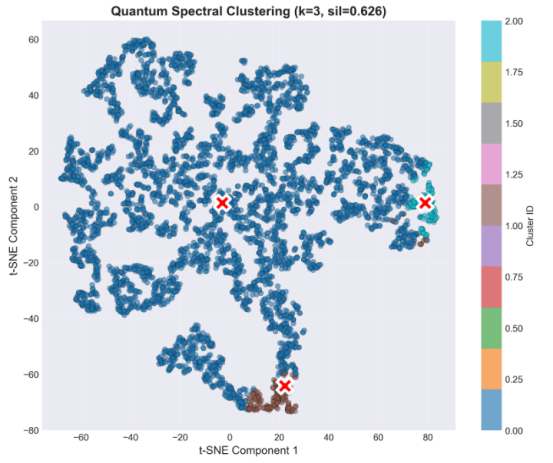


Fig. 9: t-SNE visualization of Quantum Spectral Clustering for $k = 3$ (Silhouette = 0.6263). A third cluster emerges distinctly, highlighting finer aerosol structure.

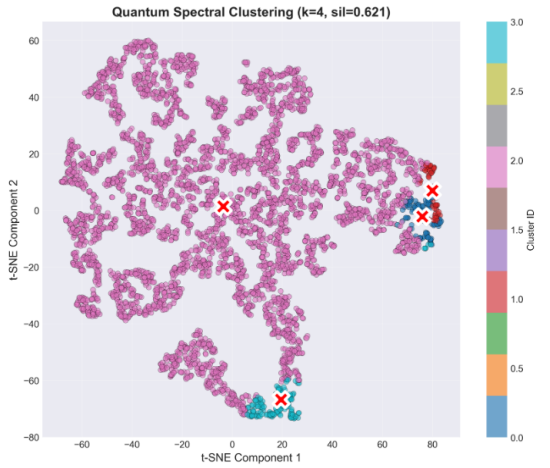


Fig. 10: t-SNE visualization of Quantum Spectral Clustering for $k = 4$ (Silhouette = 0.6211). Additional subdivision appears within dense regions, reflecting enhanced nonlinear separability.

E. Physical Interpretation and Implications

All three analytical pipelines — spectral clustering, latent-space balancing, and kernel-based clustering — produced mutually consistent aerosol regimes. The spectral signatures across EAE, AAE, SSA, and AOD ratios matched well with physically known aerosol types such as dust, biomass-burning aerosols, and mixed/urban influences. Seasonal variations in aerosol clustering aligned strongly with expected meteorological and source-driven processes in Asia, including monsoon-driven removal, spring dust storms, and post-monsoon burning. The latent-space augmentation experiment demonstrated that cluster discovery can be significantly improved by balancing under-represented aerosol conditions, with large gains in all clustering metrics. The quantum kernel results suggest that quantum affinity structures may offer advantages in identifying cluster patterns that classical kernels struggle to capture, particularly in high-dimensional but well-structured latent spaces. These benefits, however, are sensitive to latent dimensionality and kernel parameter tuning, indicating avenues for future work. Overall, the combined approach provides a robust framework for large-scale aerosol regime identification, with applications in satellite validation, air quality modeling, and long-term climate studies.

V. Conclusion

This pipeline provides a reproducible and robust framework for analyzing large aerosol datasets by integrating deterministic preprocessing, a compact autoencoder for latent space construction, and both classical and quantum-inspired kernel methods for clustering. Its design emphasizes stability and interpretability through adaptive kernel tuning, density estimation, and extensive diagnostics. Although the quantum component is implemented as a classical proxy, it offers a practical way to assess how quantum-inspired similarity measures compare with traditional RBF kernels. Overall, the

workflow effectively enhances feature structure, supports fair comparisons across clustering methods, and produces fully reproducible artifacts, making it well suited for future hyperparameter exploration or integration of genuine quantum kernels..

This research highlights the necessity of empirical analysis when selecting object detection models for real-world applications. It provides significant recommendations for developing more effective traffic surveillance systems, city planning software, and aerial observation platforms, where accurate vehicle detection in satellite imagery is crucial for decision-making and resource optimization.

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